

Analyzing the Synergistic Effects of TSMO Strategies for Quicker Speed Recovery after Incidents

Francisca Kasubi¹; H. M. Nayem²; and Priyanka Alluri, Ph.D., P.E.³

¹Graduate Research Assistant, Dept. of Civil and Environmental Engineering, Florida International Univ., Miami, FL. Email: fkasu001@fiu.edu

²Graduate Research Assistant, Dept. of Civil and Environmental Engineering, Florida International Univ., Miami, FL. Email: hnaye001@fiu.edu

³Associate Professor, Dept. of Civil and Environmental Engineering, Florida International Univ., Miami, FL. Email: palluri@fiu.edu

ABSTRACT

Transportation Systems Management and Operations (TSM&O) strategies, including Dynamic Message Signs (DMSs), Road Ranger Service Patrols (RRSPs), Severe Incident Response Vehicles (SIRVs), and Express Lanes (ELs), aim to optimize traffic flow, enhance safety, and expedite recovery after incidents. This study evaluated the combined impact of TSM&O strategies on traffic speed recovery along South Florida's I-95 corridor from 2020 to 2022. Using speed exceedance probability analysis, data from SunGuide, the National Oceanic and Atmospheric Administration (NOAA), and the Regional Integrated Transportation Information System (RITIS) were integrated to assess speed recovery patterns across various time intervals and incident severities. Results showed that combined DMS, RRSP, SIRV, and ELs deployment achieved the highest speed recovery probabilities, with over 70% of incidents recovering within 5 min for minor and 15 min for major incidents. Findings highlight the benefits of coordinated TSM&O deployment for faster traffic recovery, reduced congestion, and improved network performance, offering a framework for adaptive traffic management systems leveraging real-time data.

Keywords: Traffic Incident Management, TSM&O Strategies, Speed Exceedance Probabilities, Data-Driven Analysis

INTRODUCTION

Traffic incidents, from minor to severe incidents, significantly impact urban safety and mobility, leading to congestion and extended travel times (Taylor, 2008). Adverse weather conditions, such as rain and fog, exacerbate incident severity and complicate recovery efforts, underscoring the need for adaptive strategies (Kasubi et al., 2024). Efficient incident traffic management is essential for mitigating these impacts and restoring normalcy on our roads. Among the strategies employed, Transportation Systems Management & Operations (TSM&O) have emerged as a cornerstone for enhancing rapid traffic recovery. These strategies include the Road Ranger Service Patrol (RRSP), Severe Incident Response Vehicles (SIRV), Express Lanes (ELs), Ramp Metering Signals (RMSs), and Dynamic Message Signs (DMSs).

Previous studies have evaluated the impact of individual TSM&O strategies on traffic flow and safety (Alluri, Geedipally, et al., 2022; Alluri, Salum, et al., 2022; Gan et al., 2011; Haule et al., 2021; Kadeha et al., 2020; Salum et al., 2020, 2021). However, the determination of the most effective combination of these TSM&O strategies to tackle the unpredictable nature of traffic

incidents remains a complex challenge. Recent progress in traffic analytics, particularly the development of speed exceedance probabilities, provides a new lens through which the efficacy of TSM&O strategies can be quantitatively assessed. This metric offers insights into the likelihood of traffic speeds rebounding to normal levels at various intervals post-incident, presenting a novel approach to evaluating and optimizing traffic management strategies.

This study analyses the speed exceedance probabilities across various TSM&O scenarios along the Interstate 95 (I-95) corridor passing through Broward, Palm Beach, and Treasure Coast counties in Florida Department of Transportation (FDOT) District Four (D4). By examining different combinations of RRSP and SIRV deployments, the operational dynamics of ELs, and the timing of post-incident clearance, the study aims to identify the most suitable strategies for expediting the normalization of traffic speeds. The constant operation of DMSs across all scenarios ensures a consistent baseline for assessing the impact of variable strategies.

The findings aim to shed light on the synergistic effects of RRSP and SIRV deployment in conjunction with EL management on enhancing speed recovery probabilities. Integrating these insights into traffic management systems, such as the DSS Lite, underscores our contribution to the field. By offering a data-driven framework for incident management, this research not only addresses the immediate challenges posed by traffic incidents but also paves the way for building a more resilient and responsive transportation network.

METHODOLOGY

This study focused on the TSM&O strategies implemented along the I-95 corridor within FDOT D4. The study corridor extends over 142 miles of the I-95 Mainline and includes 12 miles of 95Express lanes. The data and TSM&O strategies analyzed span a period of three years, from January 1st, 2020, to December 31st, 2022.

Study Corridor

The TSM&O strategies analyzed in this report include DMSs, RRSP, SIRV, and ELs. The distribution of these TSM&O strategies varies along the corridor. Strategies such as DMSs, RRSP, and SIRV are distributed throughout, while ELs are limited to specific sections. RMSs were excluded from this analysis due to their limited deployment, which does not provide a sufficient data set for evaluation. Consequently, for analytical purposes, the corridor has been logically grouped into four groups. These groups are sequentially labeled according to their geographical location from the beginning to the end of the corridor's boundary.

Northbound & Southbound Direction:

- Group 1 (Sebastian Blvd to Glades Rd): Includes DMSs, RRSP, and SIRV.
- Group 2 (Glades Rd to Commercial Blvd): Includes DMSs, RRSP, SIRV, and ELs.
- Group 3 (Commercial Blvd to Sunrise Blvd): Includes DMSs, RRSP, and SIRV.
- Group 4 (Sunrise Blvd to Hallandale Beach Blvd): Includes DMSs, RRSP, SIRV, and ELs.

Figure 1 shows the study corridor along with the strategic locations of these TSM&O strategies.

Data Cleaning and Processing

Traffic speed data were sourced from Regional Integrated Transportation Information System (RITIS) detectors along I-95. TSM&O operation records included incident reports from

SunGuide™ detailing the nature of incidents and response actions, as well as DMS & ELs operations data. The dataset underwent extensive cleaning to ensure accuracy and reliability. This involved removing erroneous or incomplete entries and focusing on precise locations and times. Special attention was given to the incident and DMS data to focus on messages that directly impacted traffic flow, such as lane blockage information. Generally, the analysis focused on all incidents and DMSs containing lane blockage information. Redundant entries without critical temporal or spatial information were excluded.



Figure 1: Study Corridor

Data Integration

Several past studies have evaluated these strategies independently and are proven to be effective in improving the operational and safety performance of the freeway network (Alluri et al., 2020; Alluri, Salum, et al., 2022). However, it is unclear how these strategies work together in synergy. This study effort used a data-driven approach to uncover the synergistic benefits of TSM&O strategies along the I-95 corridor in FDOT D4. The data integration procedures involved:

- Using ArcGIS to spatially map DMS messages and incidents containing the presence or absence of RRSP and SIRV.
- Employing R studio for processing and linking data based on proximity and timing.
- Incorporating ELs status during incidents.

Speed data were recorded at three distinct intervals: pre-incident, during the incident, and post-incident. The pre-incident speed established a baseline for regular traffic conditions, absent of any disruptions. The speed captured during the incident provided insights into the immediate impact on traffic flow. The post-resolution speed data offered an indication of the recovery efficiency, reflecting the restoration of normal traffic conditions.

Speed Probability Analysis

Speed probability analysis was crucial for understanding the impact of TSM&O strategies on traffic speed in the aftermath of incidents. This method aligned with the study objectives by offering insights into traffic flow dynamics under different management conditions.

The analysis involved several vital steps.

- i. Data preparation: TSM&O and speed data were imported and cleaned.
- ii. Scenario creation and analysis
 - Scenarios were defined based on specific conditions (e.g., absence/presence of RRSP, SIRV, and ELs during the incident)
 - Probabilities were calculated to determine the likelihood of traffic speed exceeding the speed observed 5 minutes before the incident. These calculations were made for various time intervals after the incident and after the clearance of the incident (5, 10, 15, 20, 25, 30 minutes before and after the incident occurred and after clearance).
 - The probability was calculated as the mean of instances where the speed at a given time interval exceeds the speed 5 minutes before the incident.
 - This approach provided insights into how traffic speed changes in response to an incident under various traffic management conditions.
- iii. Visualization: Line plots displayed speed exceedance probabilities over time, with clear annotations for precise interpretation.
- iv. Comparative analysis: Scenarios involving various combinations of TSM&O strategy deployments were examined to assess their impact on traffic speed recovery.

RESULTS

The results show the exceedance probabilities for speed recovery on the I-95 corridor section after an incident occurred, comparing different TSM&O strategies. The probabilities are

measured at various time intervals after the incident occurred and after the incident has been cleared for both minor and major incidents.

Northbound

For minor incidents, the results reveal that the combined application of RRSP, SIRV, and ELs significantly enhances the speed of recovery post-incident. Notably, this combination yields the highest probability of returning to average speeds within 5 minutes after incident clearance, underscoring the synergy between these strategies in facilitating rapid traffic normalization.

For major incidents, speed recovery is naturally slower, with a noticeable difference in exceedance probabilities when compared to minor incidents. Nonetheless, the strategy involving RRSP, SIRV, and ELs remains superior, offering the best outcomes for major incident recovery. Table 1 summarises the speed recovery probabilities for the northbound direction.

Table 1: Speed Recovery Probabilities for Northbound

Incident Severity: Minor	Scenarios					
Time_Interval	RRSP	SIRV	RRSP & SIRV	SIRV & EL	RRSP & EL	RRSP & SIRV & EL
20min_after	0.46	0.48	0.48	0.49	0.47	0.49
25min_after	0.52	0.54	0.54	0.55	0.51	0.55
30min_after	0.58	0.60	0.60	0.61	0.57	0.61
5min_Cleared	0.71	0.71	0.71	0.73	0.72	0.73
10min_Cleared	0.72	0.73	0.73	0.73	0.72	0.73
15min_Cleared	0.74	0.76	0.76	0.77	0.75	0.77
20min_Cleared	0.74	0.77	0.77	0.79	0.75	0.79
25min_Cleared	0.75	0.78	0.78	0.77	0.74	0.77
30min_Cleared	0.76	0.78	0.78	0.81	0.77	0.81
Incident Severity: Major						
Time_Interval	RRSP	SIRV	RRSP & SIRV	SIRV & EL	RRSP & EL	RRSP & SIRV & EL
20min_after	0.36	0.36	0.36	0.36	0.47	0.49
25min_after	0.40	0.39	0.39	0.40	0.51	0.40
30min_after	0.45	0.43	0.43	0.44	0.57	0.44
5min_Cleared	0.72	0.72	0.72	0.73	0.72	0.73
10min_Cleared	0.71	0.70	0.70	0.72	0.72	0.72
15min_Cleared	0.74	0.72	0.72	0.74	0.75	0.74
20min_Cleared	0.74	0.73	0.73	0.75	0.75	0.75
25min_Cleared	0.74	0.74	0.74	0.75	0.74	0.75
30min_Cleared	0.77	0.76	0.76	0.77	0.77	0.77

Figure 2 shows a series of line graphs representing the probabilities of speed exceedance for different scenarios involving RRSP, SIRV, and the status of ELs after an incident has been cleared. Each line graph corresponds to a specific combination of the status of TSM&O strategies.

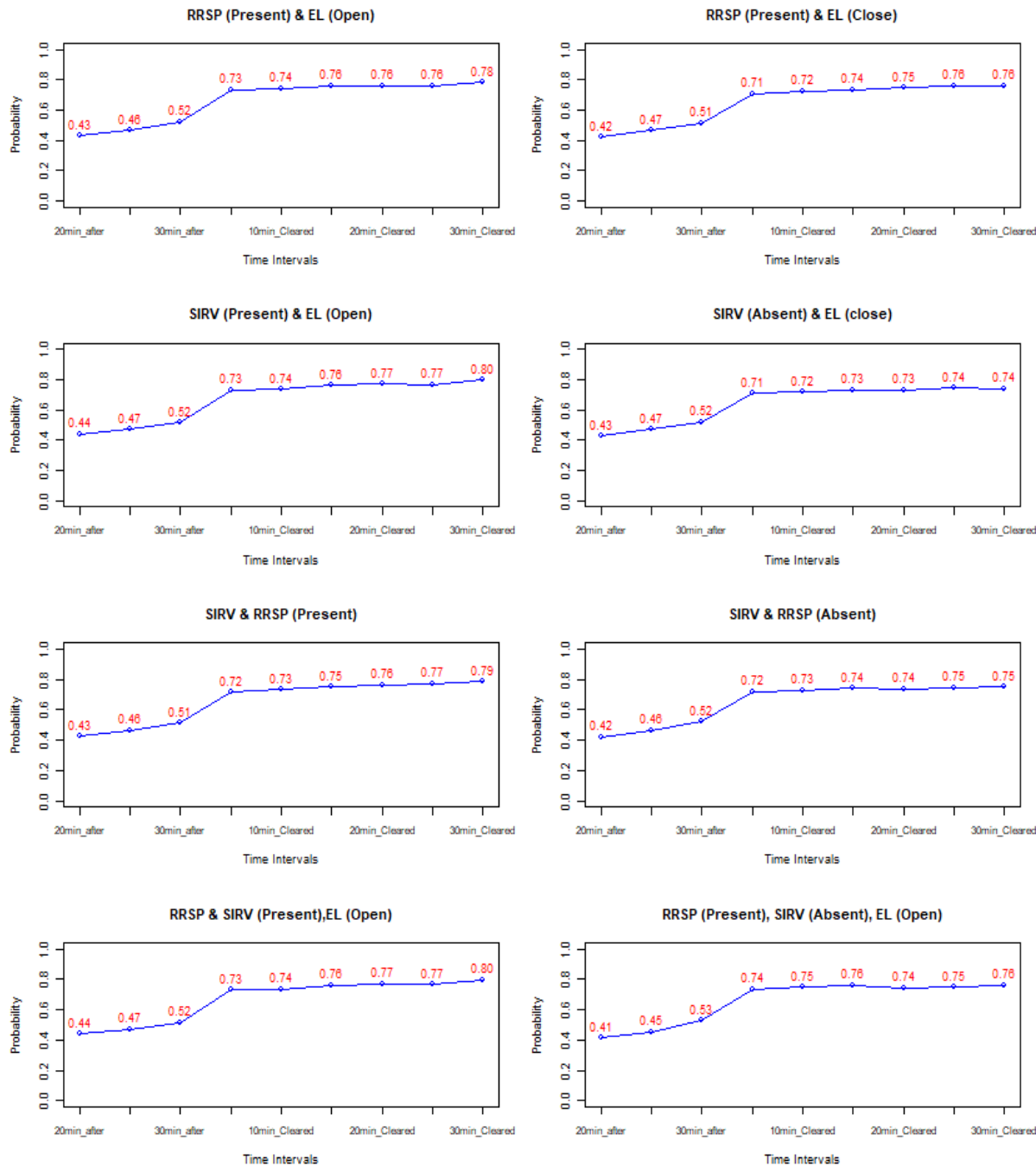


Figure 2: Speed Exceedance Probability Scenarios Comparison NB

When RRSP is present, and ELs are open, the traffic speed recovery shows promise, with the probability sitting at 0.43 twenty minutes after the incident occurred and surging to 0.73 five

minutes post-clearance. This reflects how ELs, alongside RRSP, significantly boost the recovery process. In contrast, when RRSP is still present, but the ELs are closed, the probability of speed recovery increases to 0.71 five minutes after clearance. The closure of ELs slightly hinders the quick recovery of normal traffic flow.

With SIRV Present and ELs open, the probability of speed recovery is significantly greater, underscoring the effectiveness of SIRV in conjunction with open ELs in enhancing traffic flow post-incident. On the flip side, with SIRV absent and ELs closed, the graphs indicate the lowest speed recovery probabilities, signaling this as the least effective scenario for traffic management post-incident.

Scenarios where both SIRV and RRSP are present show there's a solid recovery trajectory, with the probability at 0.43 twenty minutes after the incident occurred, escalating to an impressive 0.79 thirty minutes after the road has been cleared. This highlights the synergistic effect of these strategies on traffic recovery. In contrast, when both SIRV and RRSP are absent, there's a slight decrease to 0.75 probability of speed recovery, emphasizing the critical role these services play in traffic management after incidents.

The combination of RRSP, SIRV, and ELs being open is associated with the highest speed recovery probabilities, marking it as the most effective traffic management strategy post-incident. However, when RRSP is present with SIRV absent, even with ELs Open, there's a noticeable decrease in effectiveness, pinpointing the vital contribution of SIRV when used alongside RRSP.

The best scenario for speed recovery, applicable to both minor and major incidents, involves the presence of both RRSP and SIRV with ELs open. This scenario shows the highest probability of speed exceedance, indicating a quicker return to normal traffic conditions. Even though the recovery is slower during major incidents, the integration of RRSP and SIRV with ELs open is still the optimal approach. This suggests the indispensable nature of these services in ensuring the rapid restoration of average traffic speeds following severe incidents.

Southbound

The southbound section mirrors the northbound analysis, with TSM&O strategies showing a consistent impact on speed recovery, albeit with variations in effectiveness based on the specific combinations and incident severities.

For minor incidents, TSM&O strategies have a uniform initial impact, but as time progresses, combinations, including ELs, show higher effectiveness. Post-clearance strategies involving ELs yield the highest probabilities for speed recovery.

While major incidents present a more severe initial impact on traffic flow, the recovery patterns align with those observed for minor incidents. Post-clearance, a holistic approach (RRSP, SIRV, and ELs) demonstrates the highest effectiveness for speed recovery. Table 2 summarises the probabilities of traffic speed recovery at different time intervals post-incident for both minor and major incident severities under various TSM&O strategy combinations.

Figure 3 depicts a series of line graphs, each representing the probabilities of speed exceedance for different combinations of TSM&O.

When RRSP is present, and ELs are open, the probability of speed exceedance increases over time after the incident occurs, starting from a probability of 0.41 twenty minutes after the incident to 0.53 thirty minutes after. There is a noticeable jump in the probability of speed exceedance once the incident is cleared, starting from 0.77 five minutes after clearance. The

probability continues to increase slightly with time after the incident is cleared, plateauing at 0.79 fifteen minutes after clearance and reaching 0.80 thirty minutes after clearance. In contrast, when RRSP is still present, but the ELs are closed, the probability of speed recovery increases to 0.72 five minutes after clearance.

Table 2: Speed Recovery Probabilities for Southbound

Incident Severity: Minor	Scenarios					
Time_Interval	RRSP	SIRV	RRSP & SIRV	SIRV & EL	RRSP & EL	RRSP & SIRV & EL
20min_after	0.46	0.47	0.47	0.47	0.46	0.47
25min_after	0.54	0.54	0.54	0.55	0.57	0.55
30min_after	0.58	0.60	0.60	0.66	0.62	0.65
5min_Cleared	0.73	0.73	0.73	0.75	0.77	0.75
10min_Cleared	0.75	0.75	0.75	0.77	0.78	0.77
15min_Cleared	0.74	0.74	0.74	0.76	0.77	0.76
20min_Cleared	0.75	0.75	0.75	0.77	0.78	0.77
25min_Cleared	0.77	0.77	0.77	0.79	0.80	0.79
30min_Cleared	0.77	0.77	0.77	0.78	0.79	0.78
Incident Severity: Major						
Time_Interval	RRSP	SIRV	RRSP & SIRV	SIRV & EL	RRSP & EL	RRSP & SIRV & EL
20min_after	0.33	0.33	0.33	0.33	0.46	0.47
25min_after	0.36	0.36	0.36	0.35	0.57	0.35
30min_after	0.42	0.41	0.41	0.42	0.62	0.42
5min_Cleared	0.71	0.73	0.73	0.77	0.77	0.77
10min_Cleared	0.73	0.75	0.75	0.78	0.78	0.78
15min_Cleared	0.75	0.77	0.77	0.80	0.77	0.80
20min_Cleared	0.75	0.77	0.77	0.79	0.78	0.79
25min_Cleared	0.76	0.76	0.76	0.77	0.80	0.77
30min_Cleared	0.77	0.78	0.78	0.79	0.79	0.79

SIRV (Present) & ELs (Open) vs. SIRV (Absent) & EL (Close): Recovery is quicker when the SIRV is present and the ELs are open. This indicates that both the specialized capabilities of the SIRV and the accessibility of ELs are critical factors in improving incident management.

SIRV & RRSP (Present) vs. SIRV & RRSP (Absent): The combined deployment of SIRV and RRSP leads to superior recovery speeds compared to scenarios where both are absent. It seems that collaboration of the strategies leads to more efficient incident handling.

RRSP & SIRV (Present), EL (Open) vs. RRSP (Present), SIRV (Absent), EL (Open): Having both RRSP and SIRV available, along with open ELs, enhances recovery speed more than just having RRSP with ELs open. This implies that the SIRV provides additional benefits in managing incidents beyond what the RRSP can accomplish on its own, even when ELs are open.

For minor incidents, the best scenario for speed recovery appears to be when RRSP, SIRV, and EL are all deployed, as indicated by the higher recovery speeds across all time intervals post-

incident. For major incidents, the same seems to hold; the combination of RRSP, SIRV, and EL open presents the highest recovery speeds. However, it's notable that the difference between scenarios with and without SIRV is more pronounced in major incidents, emphasizing the importance of SIRV in more severe situations.

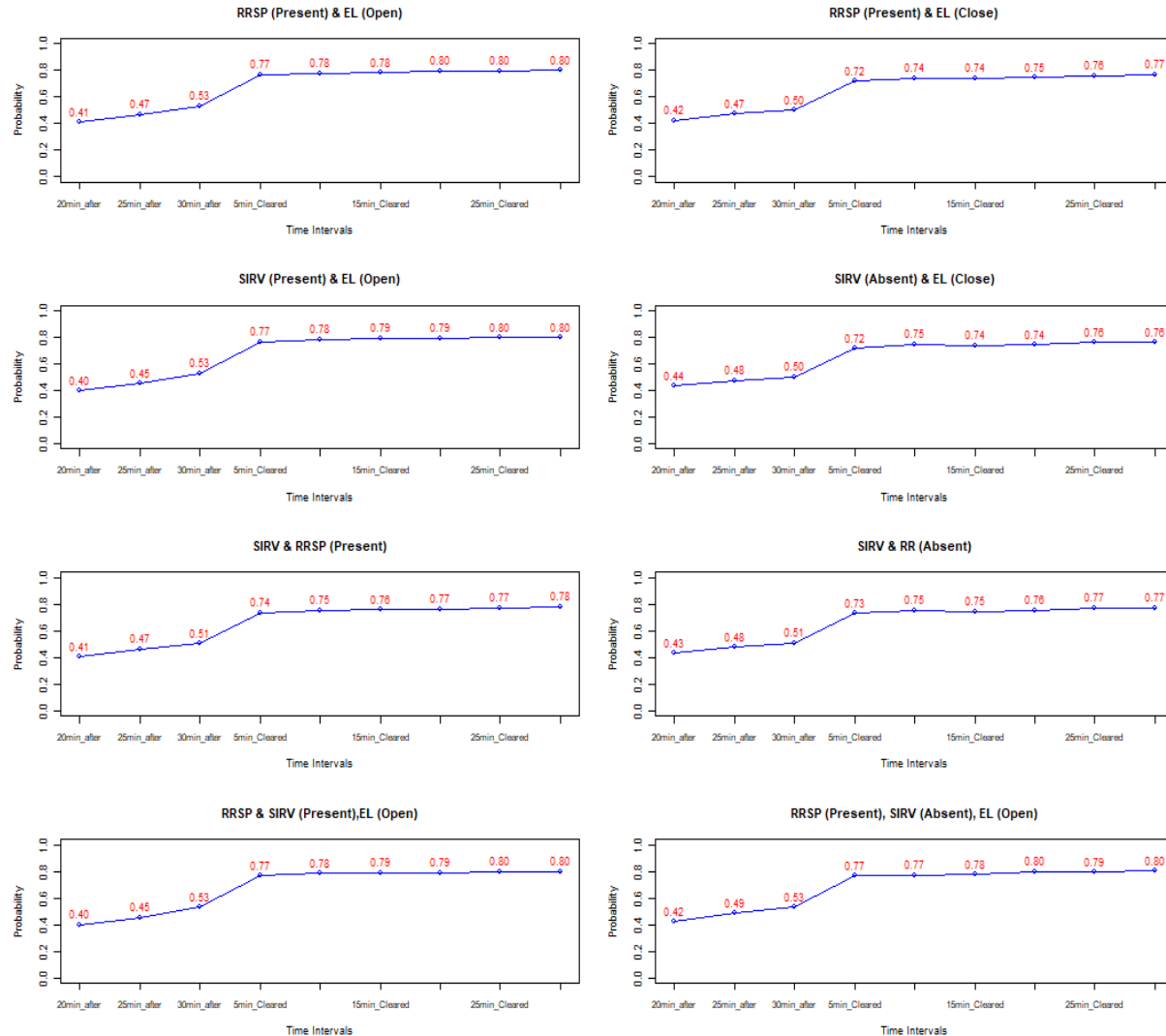


Figure 3: Speed Exceedance Probability Scenarios Comparison SB

The steady increase in the probability of speed exceedance after the incident clearance indicates the effectiveness of the TSM&O strategies in managing the incident and aiding the recovery of normal traffic flow. The trend suggests that within half an hour post-clearance, the traffic conditions significantly improve and stabilize.

CONCLUSIONS AND RECOMMENDATIONS

Upon evaluating the impact of various TSM&O strategies on the I-95 corridor, the study has yielded conclusions and formulated actionable recommendations.

Conclusions

The deployment of TSM&O strategies collectively contributes to enhanced mobility and safety. Notably, the strategic combination of RRSP, SIRV, and ELs has been effective in accelerating traffic recovery after disruptions caused by both minor and major incidents. This integrated approach consistently resulted in the highest probabilities of speed exceedance, indicating a faster return to regular traffic flow.

While minor incidents naturally exhibit a faster recovery trajectory, the data indicates that major incidents also show an improved recovery when managed under the combined TSM&O strategy. The consistency of this pattern underscores the effectiveness of utilizing RRSP, SIRV, and ELs in tandem to mitigate the impact of traffic incidents.

Recommendations

In light of these findings, it is recommended that traffic management authorities implement an integrated TSM&O strategy approach, particularly during peak hours or when incidents are more likely to occur. Encourage ELs to remain open during and after traffic incidents to ensure that there is additional capacity for traffic to recover its average speed more quickly.

Furthermore, it's essential to inform the public about the benefits of these TSM&O strategies and how they contribute to reducing traffic congestion post-incidents, possibly encouraging public support and understanding during traffic disruptions. Continue monitoring traffic data to evaluate the effectiveness of these strategies over time and make adjustments as necessary.

By adopting these recommendations, traffic management authorities can enhance the resilience of the I-95 corridor against disruptions and improve overall traffic flow and safety.

ACKNOWLEDGMENTS

This project was partially funded by the Florida Department of Transportation (FDOT). The authors thank the FDOT D4 and RITIS for providing access to the data. The opinions, findings, and conclusions expressed in this publication are those of the author(s) and do not necessarily reflect the views of any other Organization or Higher Education Institution.

REFERENCES

- Alluri, P., Geedipally, S., Salum, J., Kadeha, C., Le, M., Fitzpatrick, K., and Lord, D. (2022). Developing Safety Performance Function (SPF) and Crash Modification Factor (CMF) for Managed Lanes Separation Treatments (Issue April). <https://rosap.ntl.bts.gov/view/dot/70408>.
- Alluri, P., Salum, J., Kitali, A., Haule, H., and Angel, M. (2022). Performance evaluation of Connected Vehicle (CV) and Transportation Systems Management and Operations (TSM&O) projects in Florida. February. <https://rosap.ntl.bts.gov/view/dot/66067>.
- Alluri, P., Sando, T., Kadeha, C., Haule, H., Salum, J. H., Ali, M. S., Kodi, J. H., and Kitali, A. E. (2020). Developing Florida-specific mobility enhancement factors (MEFs) and crash modification factors (CMFs) for TSM&O strategies: Final Report (Issue March 2020). <https://fdotwww.blob.core.windows.net/sitefinity/docs/default-source/research/reports/fdot-bdv29-977-46-rpt.pdf>.

- Gan, A., Liu, K., Alluri, P., Zhu, X., and Robbins, C. (2011). Integrated Database and Analysis System for the Evaluation of Freeway Corridors for Potential Ramp Signaling. https://rosap.ntl.bts.gov/view/dot/23575/dot_23575_DS1.pdf.
- Haule, H. J., Ali, M. S., Alluri, P., and Sando, T. (2021). Evaluating the effect of ramp metering on freeway safety using real-time traffic data. *Accident Analysis and Prevention*, 157(May), 106181.
- Kadeha, C., Alluri, P., and Sando, T. (2020). Quantifying the mobility benefits of express lanes using real-time traffic data. *Transportation Research Record*, 2674(11), 414–423.
- Kasubi, F., Mdimi, O., Nayem, H., Mwambeleko, E., Shita, H., and Venkata, S. (2024). Deciphering Seasonal Weather Impacts on Crash Severity: A Machine Learning Approach. *Institute of Transportation Engineers. ITE Journal*, 94(9), 39–46. <https://www.ite.org/pub/?id=0B9B4AF5-CCB4-4126-B922-E2531C05BAED>.
- Salum, J. H., Kitali, A. E., Sando, T., and Alluri, P. (2021). Evaluating the impact of Road Rangers in preventing secondary crashes. *Accident Analysis and Prevention*, 156(December 2020), 106129.
- Salum, J. H., Sando, T., Alluri, P., and Kitali, A. (2020). Impact of Freeway Service Patrols on incident clearance duration: Case Study of Florida's Road Rangers. *Journal of Transportation Engineering, Part A: Systems*, 146(9), 04020094.
- Taylor, M. A. P. (2008). Critical transport infrastructure in urban areas: Impacts of traffic incidents assessed using accessibility-based network vulnerability analysis. *Growth and Change*, 39(4), 593–616.